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Bazak Optimization Validation Report

Client Version

Date	March 3, 2026
Experiment ID	`55a4bf49fd4730a3bb8e9bb4722f8dea`
Run ID	`06ab71ea-706d-4554-91df-40d6e0e552f6`
Optimization Method	Grid search, 4 model configurations, 3 evaluation examples each

1 Executive Summary

We completed an independent validation of the Bazak optimization results across all available system outputs and found no inconsistencies in the reported metrics.

Outcome

- Automated checks: **130**
- Passed: **130**
- Failed: **0**
- Overall conclusion: The reported results are internally consistent and can be shared.

2 What Was Validated

We compared the same run across four independent data outputs:

- Frontend CSV export (`configuration_runs_...csv`)
- SDK output (`examples/bazak/results.json`)
- SDK trial stream log (`trials_v2.jsonl`)
- SDK per-trial detail files (`trial_trial_*_v2.json`)

For each trial, we validated trial identity, model mapping, status, metric parity, latency/cost values, success-rate fields, example counts, and source-to-source deep consistency between JSONL logs and per-trial artifacts.

3 Data Evidence

3.1 Trial Set

Trial ID	Model
`trial_5822ccd7098e`	gemini-3-flash-preview
`trial_94093768acc4`	gemini-2.5-flash
`trial_4ce3058b9394`	gpt-5-nano-2025-08-07
`trial_444582cb961a`	gpt-5-mini-2025-08-07

All four sources contain the same trial IDs and model assignments.

3.2 Per-Trial Results (Consolidated)

Measure definitions used in this section

Raw quality measures

- `Tool Acc`: Correct tool selection rate.
- `Param Acc`: Correct tool-parameter rate.
- `Text Acc`: Text-response correctness rate.

Derived aggregate measures

- `Accuracy`: Trial-level mean of per-example accuracy:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \text{accuracy}_i, \quad N = 3$$

  Here, accuracy_i is the example-level `accuracy` returned for each evaluation example. In this run it is **not** computed as

$$\frac{\texttt{tool_accuracy} + \texttt{param_accuracy} + \texttt{text_accuracy}}{3}.$$

  That split-metric mean is used only as a fallback when explicit `accuracy` is missing.

- `Score`: Same as `Accuracy` in this run.

Operational measures

- `Lat(ms)`: Mean latency over examples in the trial (lower is better).
- `SDK$`: Total cost across all 3 examples (lower is better).
- `FE$`: Per-example cost displayed in FE (`SDK Total Cost / 3` for this run; lower is better).

Mdl	Raw			Drv		Ops		
	TAcc	PAcc	XAcc	Acc	Scr	Lat(ms)	SDK$	FE$
M1	0.67	1.00	1.00	0.83	0.83	11286	0.0247	0.0082
M2	0.67	0.00	1.00	0.67	0.67	16323	0.0537	0.0179
M3	0.33	0.00	1.00	0.50	0.50	36680	0.0197	0.0066
M4	0.33	0.00	1.00	0.33	0.33	32653	0.0048	0.0016

Mdl map: M1=gemini-2.5-flash, M2=gemini-3-flash-preview, M3=gpt-5-mini-2025-08-07, M4=gpt-5-nano-2025-08-07.

3.3 Raw Example-Level Results (Config + Examples -> Score)

IDs			Scores						Ops		
Trl	Mdl	Ex	ExA	TA	PA	XA	Acc	Scr	Lat	SDK$	FE$
T1	M1	E1	1.00	0.67	1.00	1.00	0.83	0.83	11286	0.0247	0.0082
T1	M1	E2	1.00	0.67	1.00	1.00	0.83	0.83	11286	0.0247	0.0082
T1	M1	E3	0.50	0.67	1.00	1.00	0.83	0.83	11286	0.0247	0.0082
T2	M2	E1	0.00	0.67	0.00	1.00	0.67	0.67	16323	0.0537	0.0179
T2	M2	E2	1.00	0.67	0.00	1.00	0.67	0.67	16323	0.0537	0.0179
T2	M2	E3	1.00	0.67	0.00	1.00	0.67	0.67	16323	0.0537	0.0179
T3	M3	E1	0.00	0.33	0.00	1.00	0.50	0.50	36680	0.0197	0.0066
T3	M3	E2	1.00	0.33	0.00	1.00	0.50	0.50	36680	0.0197	0.0066
T3	M3	E3	0.50	0.33	0.00	1.00	0.50	0.50	36680	0.0197	0.0066
T4	M4	E1	0.00	0.33	0.00	1.00	0.33	0.33	32653	0.0048	0.0016
T4	M4	E2	0.00	0.33	0.00	1.00	0.33	0.33	32653	0.0048	0.0016
T4	M4	E3	1.00	0.33	0.00	1.00	0.33	0.33	32653	0.0048	0.0016

ID legend: T1=trial_94093768acc4, T2=trial_5822ccd7098e, T3=trial_444582cb961a, T4=trial_4ce3058b9394.
Example legend: E1=no-filter-single-search-trashcan-blue, E2=product-search-specific-model, E3=consultant-fridge.
Column legend: ExA=raw example accuracy, TA=Tool Acc, PA=Param Acc, XA=Text Acc, Acc=trial accuracy, Scr=trial score, Lat=latency (ms).

3.4 Numeric Consistency Bounds

Across source-to-source comparisons, the largest absolute deltas were:

- Accuracy: `0.00003333`
- Score: `0.00003333`
- Tool accuracy: `0.00003333`
- Latency: `0.00003333 ms`
- Per-example cost: `0.00001100 USD`

These are expected formatting and rounding differences only.

4 Clarifications for Client Questions

4.1 Why FE "Cost" Differs from SDK "Total Cost"

- SDK trial metrics store total trial cost across all 3 examples.
- FE "Cost" is displayed as a per-example value in this run.
- Evidence (all trials): `FE Cost ~ SDK Total Cost / 3`.

This is a representation difference, not a data-integrity issue.

4.2 Why "Best Config" is gemini-3-flash-preview while gemini-2.5-flash has Higher Overall Accuracy

- Optimization objective for this run was **tool_accuracy** (single objective).
- **Important note:** `tool_accuracy` measures only whether the correct tool was selected. It does not directly reward parameter correctness (`param_accuracy`) or final text correctness (`text_accuracy`).
- gemini-3-flash-preview and gemini-2.5-flash tied on `tool_accuracy` (0.6667).
- Because this objective produced a tie, selection fell back to deterministic tie behavior for this run and picked gemini-3-flash-preview.
- If optimizing for overall `accuracy`, gemini-2.5-flash would rank highest (0.8333).

4.3 What If We Optimized for Other Accuracy Metrics?

Objective	Top Metric Value(s)	Selected Config	Practical Meaning
Tool accuracy	0.67 (tie)	gemini-3-flash-preview (tie handled)	Best tool-choice rate only
Param accuracy	1.00	gemini-2.5-flash	Best argument and parameter correctness
Text accuracy	1.00 (all 4 models)	No unique winner; tie across all models	Non-discriminative metric for this specific dataset
Overall accuracy	0.83	gemini-2.5-flash	Best end-to-end correctness under this scorer

Because all models scored `text_accuracy = 1.0000`, selecting this metric as the primary objective would not distinguish between configurations.

4.4 Content Novelty and Content Uniqueness

Displayed values in FE:

- Content Novelty: `0.5000`
- Content Uniqueness: `0.8336`

Meaning:

- **Content Uniqueness** estimates how different each evaluation input is from other inputs in the same dataset. Higher values mean less overlap and repetition.
- **Content Novelty** estimates how far an example is from the dataset center (how atypical it is relative to the rest).

Why these are the same across all four trials:

- They are dataset-level descriptors, not model-quality metrics.

- They are derived from the evaluation examples themselves and then shown consistently across model trials for that run.

Interpretation for this run:

- `0.8336` uniqueness indicates the three examples are fairly distinct from each other.
- `0.5000` novelty is a neutral value in this run and should not be used to compare model quality.

5 Additional Verification Notes

- We independently reviewed the end-to-end metric flow across SDK, backend persistence and analytics, and frontend extraction/CSV export.
- We validated values and structural integrity (key presence and deep metric parity between JSONL and per-trial detail files).

6 Final Assessment

This optimization run shows strong internal consistency across all recorded outputs.

Client-safe conclusion: The surprising result pattern appears to be a real outcome of the configured objective and dataset behavior, not evidence of a metric-propagation bug in this run.